Applied Thermal Engineering 171 (2020) 115082

Contents lists available at ScienceDirect

Applied Thermal Engineering

journal homepage: www.elsevier.com/locate/apthermeng

Experimental and numerical study on penetration-induced internal short-circuit of lithium-ion cell

Jionggeng Wang[b,1], Wenxin Mei[a,1], Zhixian Cui[a,1], Weixiong Shen[b], Qiangling Duan[a], Yi Jin[c], Jianbo Nie[d], Yu Tian[b], Qingsong Wang[a,*], Jinhua Sun[a]

[a] State Key Laboratory of Fire Science, University of Science and Technology of China, Hefei 230026, PR China
[b] Zhe Jiang Huayun Technology Information Co.,Ltd, Hangzhou 310000, PR China
[c] China Electric Power Research Institute, Beijing 100192, PR China
[d] State Grid Zhejiang Integrated Energy Service Company, Hangzhou 310000, PR China

HIGHLIGHTS:

- The mechanism of penetration-induced internal short-circuit of LIB is analyzed.
- Penetration experiments, simulation, and separator thermal analysis are involved.
- Two types of temperature rise modes are observed for thermal runaway case.
- The Joule heat generated at short spot constitutes the most to the total heat.

ARTICLE INFO

Keywords:
Lithium-ion battery safety
Nail penetration
Internal short-circuit
Thermal runaway

ABSTRACT

Nail penetration test is an abuse test method to evaluate the thermal hazard of lithium-ion battery. The internal short-circuit is a direct cause of battery thermal runaway, while its mechanism has not been fully understood yet. In this work, the penetration-induced internal short-circuit of lithium-ion cell is studied through nail penetration experiments, numerical simulation, and cell separator thermal analysis. The results show that the nail plays a dual role in the penetration process for thermal runaway or non-thermal runaway cases, which determines the short-circuit current and heat dissipation. In addition, there are two types of temperature rise modes for thermal runaway case, the time intervals from the beginning of penetration to thermal runaway of these two modes are less than 5 s and 60 s, respectively. It can also be concluded that the cell terminal voltage is found to decay exponentially with time after penetration, while the fluctuations and rebound of terminal voltage during the nail penetration process are attributed to the partially interrupted internal short-circuit path. The results indicate that the Joule heat generated at short spot dominates the total heat generation inside lithium-ion cell with non-thermal runaway nail penetration cases.

1. Introduction

Lithium-ion battery has obtained a widespread application as an energy source in portable electronic equipment with intrinsic features such as high energy density, long cycle-life [1,2]. Lithium-ion batteries are also frequently used in electric vehicles and energy storage systems [1,3,4]. However, the application of lithium-ion batteries is now restricted by the potential vulnerability with abuse conditions. The internal short-circuit [5] of the lithium-ion cell is intrinsically more hazardous than other abuse conditions, and it is a convenient way to emulate the lithium-ion cell internal short-circuit via the nail penetration tests, which is adopted to assess the safety performance of cells before entering the market [6,7].

A considerable amount of research has been carried out on penetration-induced lithium-ion cell failures with internal short-circuit through experimental approaches. Maleki et al. [8] investigated the consequences of cell capacity as well as internal short-circuit location on thermal stability of lithium-ion cells through experiments and thermal modeling. Lamb et al. [6] studied the influences of cell constructions on the safety of commercial cells under nail penetration.

* Corresponding author.
E-mail address: pinew@ustc.edu.cn (Q. Wang).
[1] These authors contributed equally to this work.

https://doi.org/10.1016/j.applthermaleng.2020.115082
Received 8 November 2019; Received in revised form 11 February 2020; Accepted 13 February 2020
Available online 14 February 2020
1359-4311/

Nomenclature

A_{nail}	nail cross-sectional area (m^2)
c	volume-averaged lithium concentration in a phase (mol m^{-3})
c_p	specific heat capacity (J kg^{-1} K^{-1})
D	diffusion coefficient of lithium species (m^2 s^{-1})
d_{nail}	nail diameter (m)
E_{act}	activation energy (J mol^{-1})
h	convective heat transfer coefficient (W m^{-2} K^{-1})
I_s	shorting current passing through the nail body (A)
j	volumetric reaction current (A m^{-3})
k	thermal conductivity (W m^{-1} K^{-1})
L_{nail}	nail length (m)
L_{pene}	penetration length (m)
q_e	reversible entropic heat generation rate (W)
q_j	Joule heat generation rate (W)
q_r	polarization heat (W)
R	universal gas constant, 8.3143 J mol^{-1} K^{-1}
R_{ct}	shorting contact resistance (Ω)
R_{nail}	penetrated nail resistance (Ω)
r_s	radius of solid active material particles (m)
R_{st}	shorting resistance (Ω)
T	absolute temperature (K)
T_∞	ambient temperature (K)
t	time (s)
U	cell open-circuit voltage (V)
v	Nail penetration rate (mm s^{-1})

Greek symbols

α_a	anodic transfer coefficient
α_c	cathodic transfer coefficient
ε	surface emissivity
ε_s	volume fraction of electrode phase
ε_e	volume fraction of electrolyte phase
κ	ionic conductivity of electrolyte (S m^{-1})
κ_D	diffusional conductivity of a species (A m^{-1})
ρ	density, (kg m^{-3})
σ	electronic conductivity (S m^{-1})
σ	Stefan–Boltzmann constant, 5.67e − 8 W m^{-2} K^{-4}
ϕ	electrical potential in a phase (V)
Φ	generic physiochemical property

Subscripts

0	initial value
e	electrolyte phase
max	maximum value
ref	with respect to a reference state
s	solid phase

Superscripts

eff	effective
Li	lithium species

Recently, the high-speed X-ray transmission imaging [9,10] was also used to investigate the internal short circuit in-situ, the results can not only be indicated by smoke generation, fire, or explosion, electrode changes can be observed via this new system. For example, Finegan et al. [11] studied the variability of the thermal and internal structural behavior of 18,650 cells during the nail penetration tests by this method, and the propagation of thermal runaway from the point of initiation was captured. Nonetheless, the nail penetration test is resources consuming and has poor repeatability, especially the experimental observations provide limited insight into the nail penetration mechanism, including heating mode, temperature distribution, electrochemical-thermal coupling, and the internal short-circuit process [12,13].

Consequently, it is necessary to combine the experimental and numerical methods to facilitate the understanding of the penetration-induced internal short-circuit behavior of lithium-ion cell. Nail penetration process is a highly dynamic electrochemical-thermal process, which can be accurately characterized by electrochemical-thermal models [14]. Santhanagopalan et al. [15] developed an electrochemical-thermal model to study the behavior of lithium-ion cell under the internal short-circuit, and they first proposed the four different types of internal short-circuits. Zavalis et al. [16] developed a 2D electrochemical-thermal model to investigate the electrochemically induced heating during three short-circuit scenarios of a prismatic battery cell. However, the cell geometry was simplified. Afterwards, Fang et al. [17] developed a 3D electrochemical-thermal model to study the internal short-circuit for a 1 Ah lithium-ion cell. The temperature curves obtained at the short spot of Anode-Aluminum shorting and Anode-Cathode shorting were compared. Feng et al. [18] studied the relationship between voltage, current, and temperature data and the internal short-circuit status through a 3D electrochemical-thermal-internal short-circuit coupled model. Zhao et al. [12,19] performed nail penetration tests as well as electrochemical-thermal modeling on batteries with different capacities and electrode parameters. They proposed an experimental method to measure the shorting contact resistance. Xu et al. [20] established a 1D electrochemical-3D thermal coupled model to predict the short-circuit behavior in a cylindrical battery cell, design parameters affecting voltage drop and temperature rise behaviors of cell are studied. Besides, one-point short-circuit and multi-point short-circuit of lithium cell are investigated. Liang et al. [13] developed a 3D-layered electrochemical-thermal coupled model based on multilayer construction of a cell to investigate the nail-penetration process in a lithium-ion cell. Noelle et al. [21] investigated the dynamic contributions of internal resistance to Joule heat generation in the earliest stages of battery failure induced by mechanical abuse.

Among the previous efforts, they are lack of the comprehensive analysis and mechanism on penetration-induced internal short-circuit behaviors, and not combine the numerical and experimental methods to investigate the internal short-circuit. In this paper, the lithium ion cell nail penetration tests with two modes are first conducted at different state of charge (SOC), nail diameter, and nail penetration rate to analysis the penetration behavior. Subsequently, a 3D electrochemical-thermal coupled model is presented to describe the behavior of cell under non-thermal runaway nail penetration case, which can help to interpret the experimental phenomenon and reveal the mechanism of nail penetration induced internal short-circuit.

2. Experimental

2.1. Lithium-ion cell nail penetration tests

The commercial 1 Ah $Li(Ni_xCo_yMn_z)O_2$ (NCM)/graphite pouch cells with tri-layer separator of polypropylene/polyethylene/polypropylene (PP/PE/PP) and aluminum shell are used. The specification of the cell is listed in Table1. All of the cells are pre-cycled three times with 0.5C rate at room temperature by using Neware battery testing system (BTS-610). Subsequently, the nail penetration tests are conducted: the tungsten steel nail penetrates the geometric center of the cell in a direction perpendicular to the electrode, then the nail remains in this position after a complete penetration (the nail has gone through the entire cell

Table 1
The specification of the cell and heat shield.

Specification	Value
Cell	
Nominal voltage	3.7 V
Rated capacity	1 Ah
Cut-off voltage	2.75 ~ 4.2 V
Weight	19.7 g
Dimension	49 × 34 × 5.5 mm^3
Charge-discharge current	0.5 C (0.5 A)
Heat shield	
Thermal conductivity	0.1 W m^{-1} K^{-1}
Thickness of the heat shield	1 mm

thickness and come out the other side of the cell face). The ventilating fan inside the nail penetration machine is turned on in the middle stage of tests to exhaust poisonous smoke.

Two categories of nail penetration tests are performed in this study, type 1 (with heat shield) and 2 (without heat shield), which is distinguished by whether there is a heat shield placed between cell and nail penetration machine or not. The heat shield is a material with low thermal conductivity to slow down the heat transmission, so that increase the probability of thermal runaway. The specification and layout of the heat shield are displayed in Table 1 and Fig. 1, respectively. Table 2 shows the experimental scheme of the cell nail penetration tests. The nail penetration tests are performed under different conditions of SOC (%), nail diameter (mm), and nail penetration rate (mm s^{-1}), which is based on the variable-controlling principle. Apart from video monitoring of the entire experimental process, temperature evolutions are continuously measured at four temperature measurement locations (TML) using four K-type thermocouples. In addition, the terminal voltage of cell is recorded instantaneously in nail penetration test type 2. The test setup of lithium-ion cell nail penetration and thermocouples layout are shown in Fig. 1.

2.2. Thermal analysis of lithium-ion cell separator

In order to clarify the mechanism of the nail penetration induced internal short-circuit, a fresh cell is disassembled to obtain its separator, and then followed by the thermal analysis of the separator which is carried out on C80 micro calorimeter. The separator is heated in an argon atmosphere at a constant heating rate of 0.8 °C min^{-1}, and the heat flow of separator is measured throughout the whole heating process. Cells after penetrating are disassembled to gain an in-depth understanding of penetration-induced thermal runaway.

3. Numerical model

3.1. Cell geometry

The geometric model is based on nail penetration test type 2 without heat shield with non-thermal runaway cases. When the cell is completely penetrated, each electrode plate is shorted independently by a part of the nail. A closed-loop current path is formed inside each electrode plate, and no current flows from the tabs to different electrode plates. The cell is composed of several stacked electrode plates, we defines one electrode plate unit includes five parts: positive current collector, cathode, separator, anode, negative current collector, all of the electrode plate units can be assumed to be identical according to the same electrochemical behavior. And given that the cell electrode thickness (μm) is much smaller than the cell length and cell width (at least mm) [14], only one electrode plate unit is used for solving the electrochemical-thermal model [22]. Fig. 2 illustrates the model geometries and the computational mesh. The whole electrode plate unit is penetrated by the nail at the beginning of the simulation but not the penetration process. All the parameters used in the model are adopted from Refs. [16,23–25] and COMSOL Material Library (2016) [26] or measured. Some of the parameters are listed in Table 3.

3.2. 3D electrochemical-thermal coupled model

The 3D electrochemical-thermal coupled model is composed of two main parts: the electrochemical part and the thermal part. The electrochemical part is based on the works of Newman et al. [27], in which Butler-Volmer equation is used to describe the electrochemical kinetics at the electrode-electrolyte interface, Fick's law is used to describe the

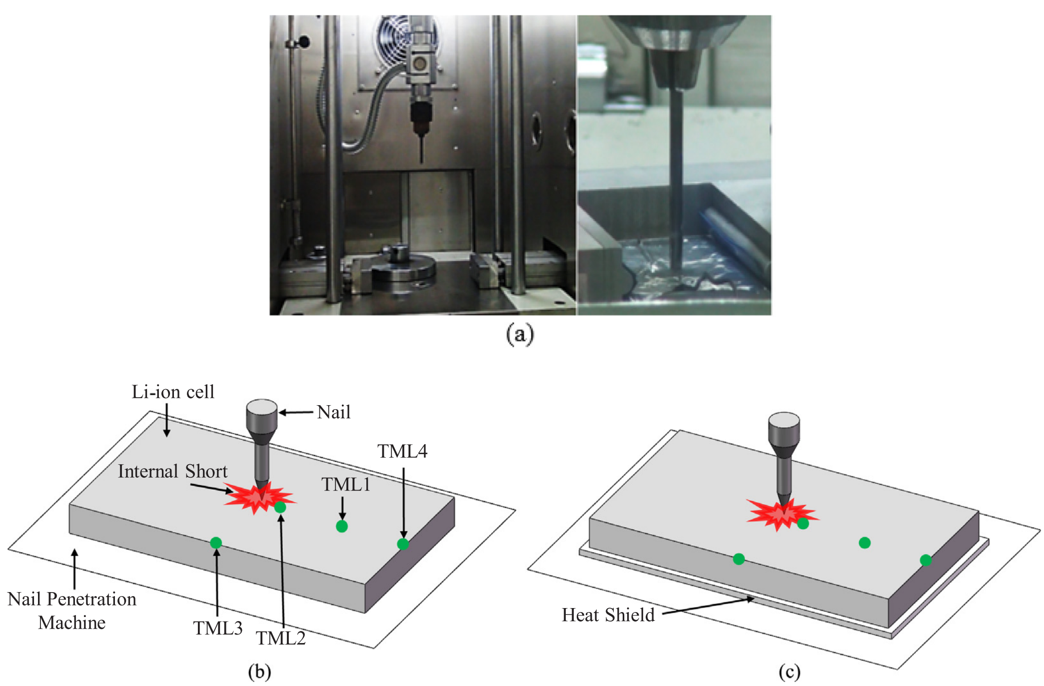

Fig. 1. Test setup and temperature measurement locations (TML) of lithium-ion cell nail penetration. (a) Photograph; (b) schematic diagram without heat shield; (c) schematic diagram with heat shield.

Table 2
Experimental conditions of test type 1 (with heat shield) and 2 (without heat shield).

Nail test type 1(with heat shield)				Nail test type 2(without heat shield)			
Experiment No.	SOC	d_{nail} (mm)	v (mm s^{-1})	Experiment No.	SOC	d_{nail} (mm)	v (mm s^{-1})
1–1	0	5	20	2–2	50	5	20
1–2	50	5	20	2–4	100	5	20
1–3	75	5	20	2–5	100	3	20
1–4	100	5	20	2–6	100	8	20
1–5	100	3	20				
1–6	100	8	20				
1–7	100	5	30				
1–8	100	5	40				

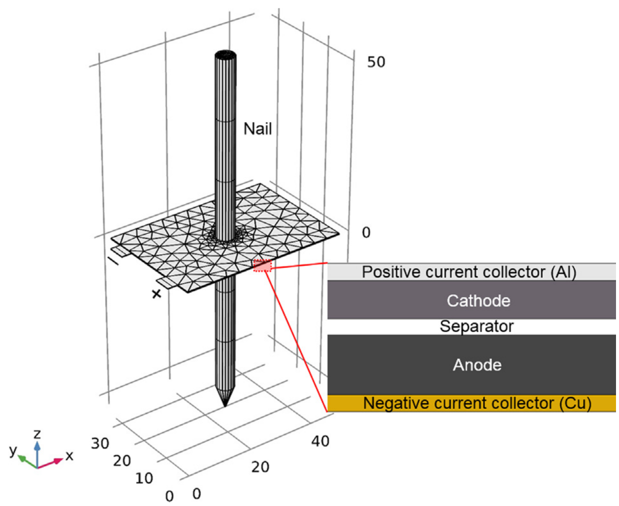

Fig. 2. Model geometries and the computational mesh for lithium-ion cell used in the nail penetration simulation.

transport of solid lithium in the solid electrode phase, and concentrated electrolyte theory for a quiescent aprotic (1:1) electrolyte is used to describe charge and mass transport in the electrolyte phase. The development of an electrochemical model has been well documented in the literature [14,16,17,27–30] and is therefore not reproduced here for brevity. The thermal part is used to determine the heat generation and temperature distribution inside the lithium-ion cell. The energy balance equation for every domain is expressed as Eq. (1) [14,16]:

$$\frac{\partial(\rho c_P T)}{\partial t} = \nabla\cdot(k\nabla T) + q_e + q_r + q_j \tag{1}$$

where ρ, c_P, and k are the volume averaged density, specific heat capacity, and thermal conductivity, respectively, T is the temperature, q_e the reversible entropic heat, q_r the polarization heat and q_j the Joule heat, side reactions are excluded in thermal model given the fact that no thermal runaway case was observed in nail penetration tests type 2 (without heat shield). q_e, q_r, and q_j are given as Eqs. (2)–(4) [14,17]:

$$q_e = j^{Li}\left(T\frac{\partial U}{\partial T}\right) \tag{2}$$

$$q_r = j^{Li}(\phi_s - \phi_e - U) \tag{3}$$

$$q_j = \sigma\nabla^2\phi_s + \kappa^{eff}\nabla^2\phi_e + \kappa_D^{eff}\nabla \ln c_e \nabla\phi_e + \sigma\nabla^2\phi_m \tag{4}$$

here j^{Li} is the reaction current density, U the cell open-circuit voltage, ϕ_s and ϕ_e the electrical potential in active material and in electrolyte, σ^{eff} the effective electronic conductivity of an electrode, κ^{eff} the effective ionic conductivity of electrolyte, κ_D^{eff} the effective ionic diffusional conductivity of electrolyte, c_e the volume averaged lithium-ion concentration in electrolyte phase and ϕ_m the electrical potential in

Table 3
Physiochemical parameters used in the model, adopted from Ref. [16,23–25] and COMSOL Material Library (2016) [26] or measured.

Parameter	Cu foil	Anode	Separator	Cathode	Al foil
Density, ρ, kg m^{-3}	8960	2270	940	2500	2700
Specific heat, c_p, J $kg^{-1}K^{-1}$	385	881	1046	1000	900
Thermal conductivity, k, W m^{-1} K^{-1}	400	1	0.15	3.4	238
Electrical conductivity, σ, S m^{-1}	5.998e7	100	N/A	100	3.774e7
Thickness, δ, µm,	25	98	13	56	25
Maximum Li^+ capacity, $c_{s,max}$, mol m^{-3}	N/A	31,507	N/A	29,000	N/A
Particle radius, r_s, µm	N/A	2	N/A	2	N/A
Electrode phase volume fraction, ε_s	N/A	0.62	N/A	0.48	N/A
Porosity, ε_e	N/A	0.31	0.4	0.29	N/A
Average electrolyte concentration, c_e, mol m^{-3}	N/A	1200			N/A
Charge-transfer coefficients, α_a, α_c	N/A	0.5, 0.5	N/A	0.5, 0.5	N/A
Li diffusion coefficient in solid, D_s, m^2 s^{-1}	N/A	1.4523e-13	N/A	5e-13	N/A
Bruggeman tortuosity exponent, p	N/A	1.5			N/A
Initial temperature, T_0, K	293.15				
Reference temperature, T_{ref}, K	293.15				
Ambient temperature, T_∞, K	293.15				
Nail diameter, d_{nail}, mm	3/5/8				
Nail length, L_{nail}, mm	107				
Penetration length, L_{pene}, µm	217				
Nail density, ρ, kg m^{-3}	1.94e4				
Nail specific heat, C_p, J kg^{-1} K^{-1}	130				
Nail thermal conductivity, k, W m^{-1} K^{-1}	180				
Nail electrical conductivity, σ, S m^{-1}	1.825e7				

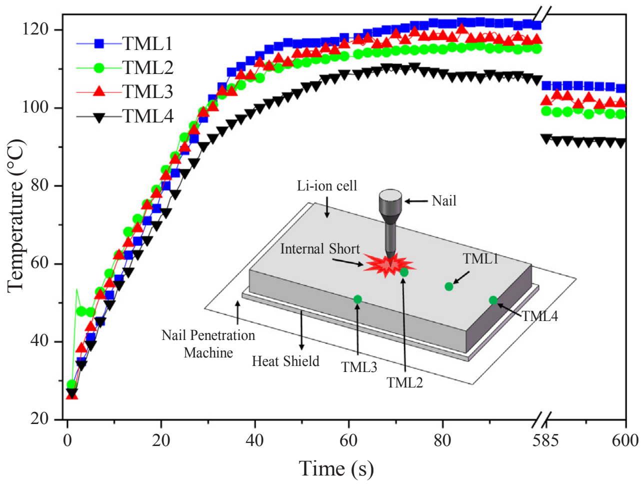

Fig. 3. Temperature *vs.* time plots for the four TMLs of a non-thermal runaway penetration case. The experimental condition is set at SOC of 100%, nail diameter of 5 mm, and the penetration rate of 20 mm s^{-1}.

metal conductor such as current collectors and the steel nail. The first term on the right-hand side of Eq. (4) represents the heat generated in solid electrode, the following two terms represent the heat generated in electrolyte, and the last term refers to the Joule heat generated in current collectors and in the steel nail, the current collectors and the steel nail only generate Joule heat.

Arrhenius equation [17,31], as shown in Eq. (5), is used to describe the temperature dependence of several electrochemical parameters, such as the exchange current density of negative electrode reaction and the diffusion coefficient of electrolyte, with which the thermal model can be coupled with the electrochemical model.

$$\Phi = \Phi_{ref}\exp\left[\frac{E_{act,\Phi}}{R}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right] \tag{5}$$

where Φ_{ref} is a generic physiochemical property at the reference state, $E_{act,\Phi}$ the activation energy of the evolution process of Φ, which is a sign of the sensitivity of parameter Φ to temperature, and T_{ref} the reference temperature.

3.3. Side reaction

Some side reaction inside the cell can be triggered at a high temperature as low as about 90 °C [32]. For example, the solid electrolyte interface (SEI) will start to decompose at 90–120 °C [33]. Subsequently, the separator will melt and further lead to the short circuit inside the cell at approximately 130–150 °C [34]. And the decomposition of cathode will be triggered at around 210 °C for NCM electrode [35] and 230 °C for Li_xCoO_2 electrode [36]. For the internal short circuit tests without heat shield studied in the present work, only in the small area around the nail site the temperature has exceeded 120 °C. Additionally, there is no thermal runaway observed in case of no heat shield as mentioned above. Therefore, no side reactions are involved in the model as the Refs. [17,19] stated.

3.4. Initial and boundary conditions

Uniform initial temperature condition is presumed, as expressed in Eq. (6).

$$T = T^0 \tag{6}$$

where T is the absolute temperature and T^0 is the initial of temperature.

The terminal voltage of cell recorded in nail penetration tests type 2 is used as the boundary condition in the numerical model, seen in Eq. (7).

$$\phi_s^- = 0,\ \phi_s^+ = TV(t) \tag{7}$$

where ϕ_s^- and ϕ_s^+ are the negative and positive current collector boundary potentials separately. TV(t) is the measured cell terminal voltage which will be discussed in section 4.

As for the thermal boundaries, heat is assumed to dissipate to the external environment in the forms of thermal convection and radiation, which is expressed as Eq. (8) [19]: the two terms on the right of the Eq. (8) represent heat convection and heat radiation, respectively.

$$-k\frac{\partial T}{\partial n} = h(T - T_\infty) + \varepsilon\sigma(T^4 - T_\infty^4) \tag{8}$$

where h is the convective heat transfer coefficient, T_∞ the ambient temperature, ε the surface emissivity and σ the Stefan-Boltzmann constant. Natural (h = 5 W m^{-2} K^{-1}) and forced (h = 55 W m^{-2} K^{-1}) convective heat transfer coefficients are applied before and after the starting of the exhaust fan, respectively. The value of ε is set at 0.8 according to the Ref. [13].

3.5. Numerical procedures

All the governing equations are solved using COMSOL Multiphysics version 5.2a (COMSOL, Inc.). The computational mesh is illustrated in Fig. 2. The mesh near the short-circuit region is refined due to its large variable gradients. A smoothed Step Function is used to ramp the conductivity of the nail from quite a low value to its full conductivity within the first 1 ms of the calculations in order to provide the model with good convergence and to induce the short-circuit. It is worth mentioning that model is only a tool to try to interpret some of the experimental results, but without claiming to be able to predict the behavior of cells in similar conditions.

Table 4
The experimental phenomenon for all of the conditions with heat shield.

Experimental condition				Thermal runaway			Phenomenon	
No.	SOC	d_{nail} (mm)	v (mm s^{-1})	Occur	Trigger time (s)	T_{max} (°C)	Gas, smoke, electrolyte boil	Burn
1–1	0	5	20	×		51.7	√	×
1–2	50	5	20	×		68.3	√	×
1–3	75	5	20	×		114.2	√	×
1–4(case1)	100	5	20	×		116.2	√	×
1–4(case2)	100	5	20	√	2.5	451.9	√	√
1–5(case1)	100	3	20	√	43	520.6	√	√
1–5(case2)	100	3	20	√	64.5	565	√	√
1–5(case3)	100	3	20	×		120.9	√	×
1–6	100	8	20	×		115.3	√	×
1–7	100	5	30	×		116	√	×
1–8(case1)	100	5	40	√	4.5	538.7	√	√
1–8(case2)	100	5	40	×		121.4	√	×

4. Results and discussion

4.1. Nail penetration tests type 1 (with heat shield)

Fig. 3 illustrates the temperature curves of the four TMLs on cell surface with a non-thermal runaway case with the SOC of 100%, nail diameter of 5 mm, and the penetration rate of 20 mm s^{-1}. Severe phenomenon such as bulging or fire is not detected throughout the penetration process, only smoking, electrolyte boiling and splashing in the short area are observed. The specific experimental phenomenon for all of the tests is summarized in Table 4. It can be noticed that temperature evolution recorded by each thermocouple follows the same trend that the temperature increases to a maximum value within about 100 s after the initiation of short and then drops gradually over time. The maximum temperature (T_{max}) captured by the four thermocouples is about 120 °C, which is below the meltdown temperature of cell separator (171.11 °C). As can be seen that there is an abrupt rise in temperature at TML2 (nail site) at the moment of penetration, which is attributed to the huge shorting current generated at the penetration moment. The T_{max} is captured at TML2 within the first 30 s after nail penetration, and a higher temperature is captured near the short-circuit area, indicating that the Joule heat generated at the shorting resistance dominates the total heat of the cell, and this Joule heat is propagated in all directions from the short-circuit area. Whereas after 30 s, the temperatures at TML1 and 3 are greater than that at TML2, which is due to the heat conduction effect of the nail.

The temperature profiles at nail site with non-thermal runaway cases for all experimental conditions are shown in Fig. 4. A sharp rise of temperature at nail site can be observed at the moment of penetration in three sub-graphs. As displayed in Fig. 4a, the temperature rising rate and the T_{max} of the cell both increase as the SOC increases, which is aroused by a greater short-circuit current and a larger Joule heat generation rate generated at a higher SOC. Fig. 4b shows the effect of nail diameter on cell temperature, it can be found that the temperature rising rate and the T_{max} of the cell both decrease as the nail diameter increases. The temperature rise is determined by both the heat generation and heat dissipation, however, in case of non-thermal runaway, the effect of the heat dissipation is larger than the heat generation, therefore, the heat dissipation dominates the temperature rise. A larger nail diameter provides larger shorting area, causing larger gross heat conducted from short spot to the nail, in other words, displaying a greater ability in heat dissipation. Therefore, the maximum temperature at nail site follows the sequence of $T_{max,3mm} > T_{max,5mm} > T_{max,8mm}$ in case of non-thermal runaway. While there displays different trend in case of thermal runaway at different nail diameter, which will be discussed in the following. Temperature curves at nail site under three different nail penetration rates are exhibited in Fig. 4c, it can be seen that the T_{max} of cell tends to increase as the nail penetration rate decreases. The nail penetration rate is assumed to have an impact on the contact resistance generated by the imperfect contact between the nail and cell body [14], which influences cell temperature rise in turn. The effect of nail penetration rate on contact resistance still needs further study.

Only fully charged cells (SOC of 100%) with nail diameter of 3 and 5 mm trigger into thermal runaway during the nail penetration tests. Fig. 5 illustrates the temperature curves at nail site during thermal

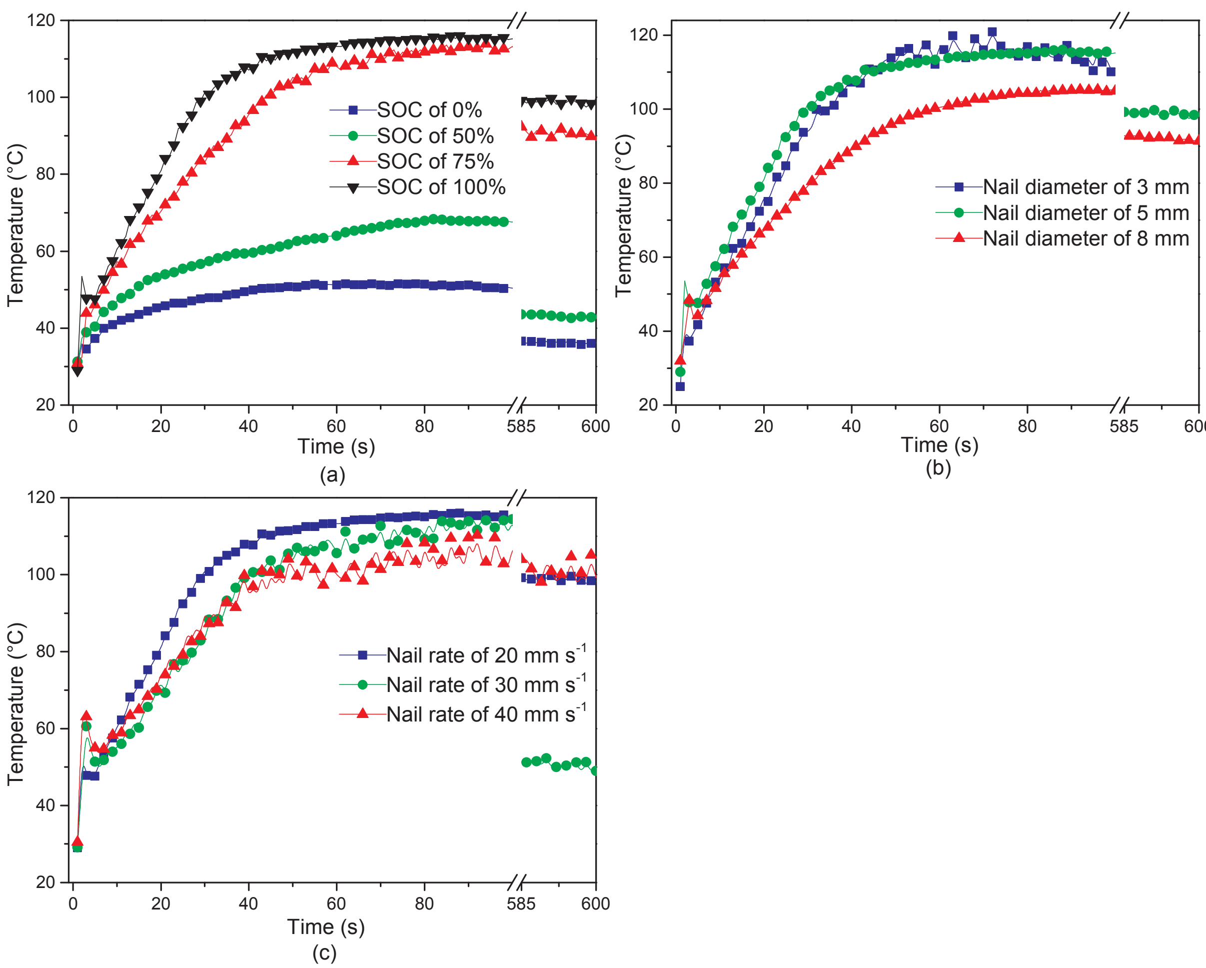

Fig. 4. Temperature profiles at nail site from nail penetration tests type 1 (with heat shield) with non-thermal runaway cases for all experimental conditions. (a) Nail diameter of 5 mm, nail rate of 20 mm s^{-1}; (b) SOC of 100%, nail rate of 20 mm s^{-1}; (c) SOC of 100%, nail diameter of 5 mm.

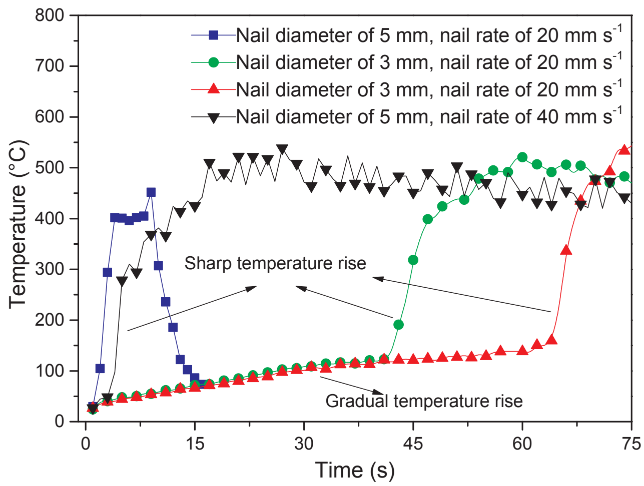

Fig. 5. Temperature *vs.* time plots at nail site during thermal runaway of the four cases. The SOC is set at 100%.

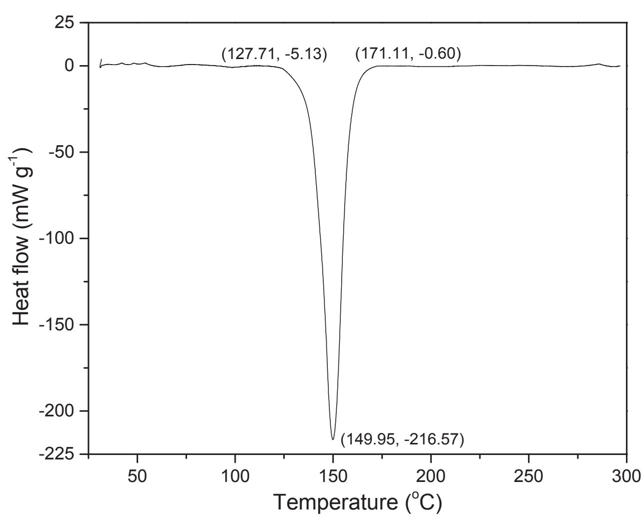

Fig. 6. C 80 micro calorimeter heat flow diagram of lithium-ion cell separator.

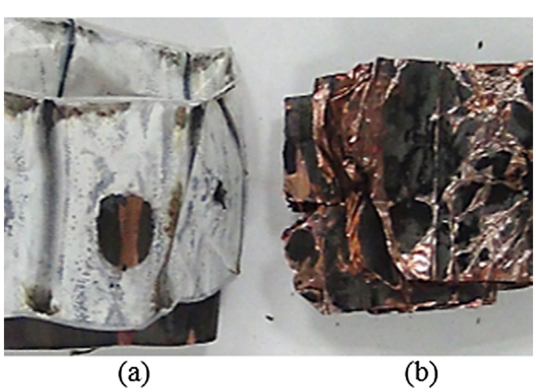

Fig. 7. Picture of jellyroll materials obtained from two penetrated lithium-ion cells. (a) Non-thermal runaway case and (b) thermal runaway case.

runaway of the four cases. A great distinction can be found in the early stage of temperature rise as noted in Fig. 5, the cell temperature undergoing a gradual rise in the early stage before a sharp rise with nail diameter of 3 mm, while the cell directly undergoing a sharp temperature rise period to a peak of over 600 °C within a few seconds after the initiation of the internal short-circuit with nail diameter of 5 mm. As listed in Table 4, the time duration from penetration to thermal runaway is relatively long (43 and 64.5 s) for experimental No.1–5 (cases1 and 2) with nail diameter of 3 mm, whereas it only lasts several seconds (2.5 and 4.5 s) for experimental No. 1–4(case2) and 1–8(case1) with nail diameter of 5 mm to undergo thermal runaway. The temperature trend displays a different regular as compared to the Fig. 4b, which shows the difference between the thermal runaway and non-thermal runaway cases. In case of thermal runaway, the effect of the heat generation is larger than the heat dissipation, thus the heat generation dominate the temperature rise. The nail with a larger diameter provides a larger short-circuit current, which consequently generates a greater total heat generation rate and thus takes less time to trigger thermal runaway. Therefore, nail diameter of 5 mm displays sharp temperature rise and earlier thermal runaway as compared to 3 mm. However, in case of 8 mm and if we continue to increase the nail diameter indefinitely, the effect of heat dissipation is larger than the heat generation, so that the thermal runaway is no longer occurred. Few smoke and electrolyte boiling in the penetration area are observed during gradual temperature rise period, while significant cell bulging and fire occurred during abrupt temperature rise period. Thermocouples dropping due to bulging of cell caused by thermal runaway led to the disagreeable occurrence of data missing (sudden drop in the temperature at about 10 s) in experimental No. 1–4 (case) with the nail diameter of 5 mm and nail rate of 20 mm s^{-1}. Examining the Table 4 can be concluded that some of the tests have thermal runaway and others not even at the similar condition. This may be caused by the inconsistencies in the cells, and the environment may not be exactly the same, this is also called "experimental contingency", which is inevitable.

4.2. Thermal behavior of the separator

Fig. 6 displays the heat flow diagram of the cell separator obtained using C80 micro calorimeter. The separator is going through an endothermic process while heating when the heat flow is negative. As shown from the Fig. 6 that the separator started endothermic process when the temperature reached 127.71 °C, and the corresponding heat absorption rate of the separator is 5.13 mW g^{-1} at that time. At a time when the temperature reaches to 149.95 °C (considered as the shutdown temperature of the separator), the heat absorption rate of the separator reached its maximum value of 216.57 mW g^{-1}. As the temperature continued to rise, separator is going to completely meltdown once 171.11 °C (considered as the meltdown temperature of the separator) is reached, the corresponding heat absorption rate dropped to 0.6 mW g^{-1}. The calculated gross heat that separator absorbed during its melting process is 210.46 J g^{-1}. Given the gross mass of separator in one cell is about 0.397 g, the heat absorbed by separator during thermal runaway process is roughly 83.55 J, which is negligible compared with the gross heat generated during a thermal runaway process (about 15 kJ).

The cells are disassembled after nail penetration tests. Fig. 7 shows the picture of jellyroll materials obtained from two penetrated cells with and without thermal runaway. As can be seen that only the separator in contact with the nail site is shrunk while the rest part maintained integrity in a non-thermal runaway case, in this case, the separator is still able to provide good insulation between anodes and cathodes after the cell is penetrated. However, the whole separator melted and the anodes and cathodes are directly contacted in a thermal runaway case, which triggered the anode–cathode electrochemical reaction and generated a large amount of heat.

4.3. Nail penetration tests type 2 (without heat shield)

None thermal runaway is occurred in the nail penetration tests type 2 (without heat shield), due to the fast heat dissipation without heat shield. A comparison of temperature at nail site and voltage curves of nail penetration tests type 2 are displayed in Fig. 8. As shown in Fig. 8a,

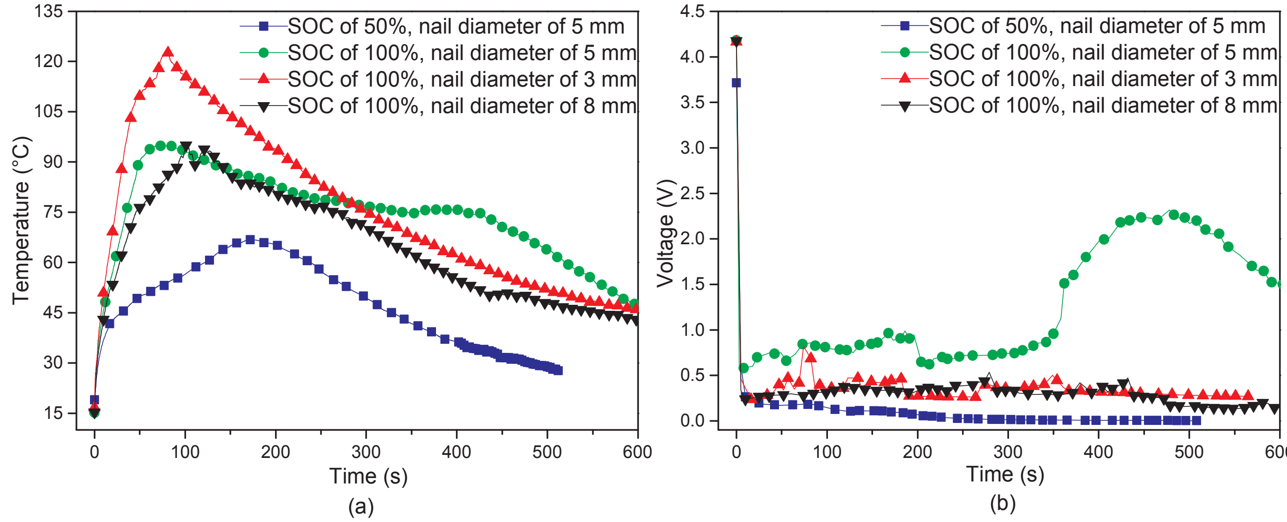

Fig. 8. Comparison of temperature at nail site (a) and terminal voltage curves (b) of nail penetration tests type 2 (without heat shield).

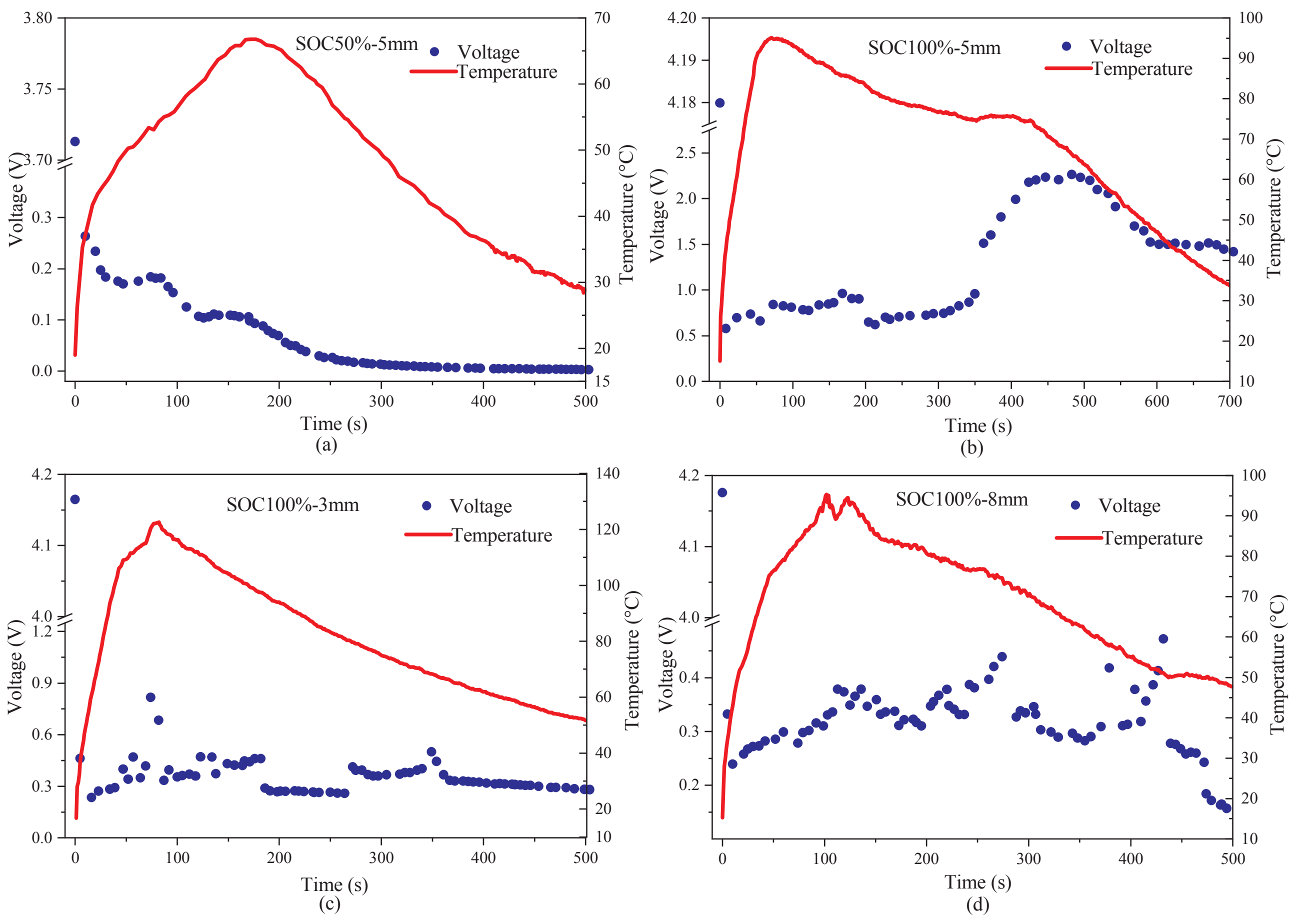

Fig. 9. The experimental results on temperature at nail site and voltage for all cases from nail penetration tests type 2 (without heat shield). (a) SOC of 50%, nail diameter of 5 mm; (b) SOC of 100%, nail diameter of 5 mm; (c) SOC of 100%, nail diameter of 3 mm; (d) SOC of 100%, nail diameter of 8 mm.

the temperature rising rate and the T_{max} at nail site both increase as the cell SOC increases or as the nail diameter decreases, which is consistent with the Fig. 4 with non-thermal runaway cases (nail penetration tests type 1 (with heat shield)).

Fig. 9 further depicts the temperature and voltage of the four cases separately. As can be seen in Fig. 9a, c d, cell terminal voltage drops significantly from its initial value to a very small value within several seconds after the penetration of the nail and then decreases with time gradually. In rare cases, cell terminal voltage can rebound back to a relatively high value as illustrated in Fig. 9b. The fluctuations and rebound of cell terminal voltage during the nail penetration process are concerned with the electrical, chemical and electrochemical processes inside the cell. The terminal voltage of cell will drop significantly once the cell is penetrated, which is attributed to the large short-circuit current, the short-circuit current will decrease as the experiment progresses. However, in some cases [37], the discharge is incomplete and

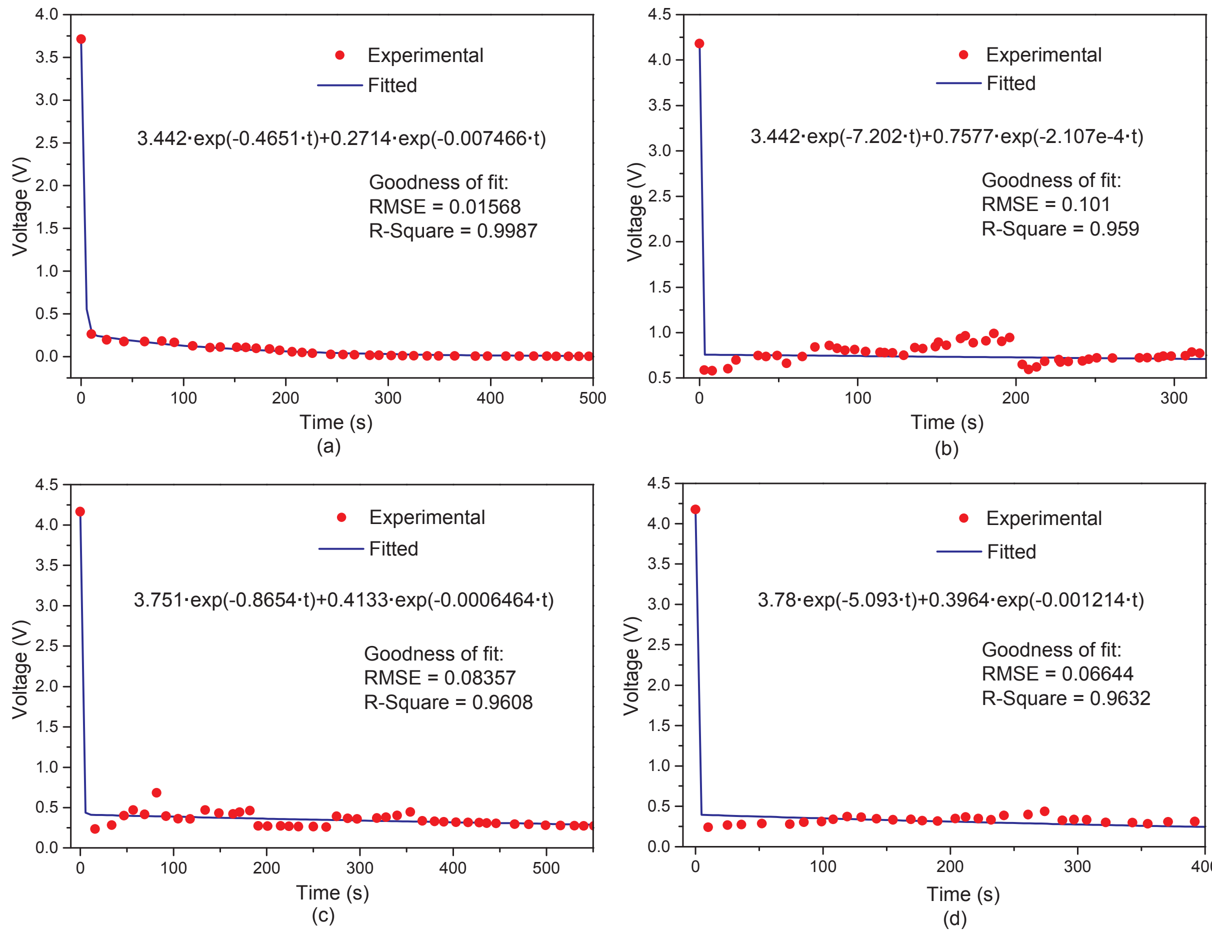

Fig. 10. Experimental terminal voltage and the corresponding fitted curves used in the model. (a) SOC of 50%, nail diameter of 5 mm; (b) SOC of 100%, nail diameter of 5 mm; (c) SOC of 100%, nail diameter of 3 mm; (d) SOC of 100%, nail diameter of 8 mm.

the internal short-circuit path is partially interrupted, thus causing the higher voltage and lower short-circuit current. It is difficult to measure the short-circuit current in the experiment, hence the simulated short-circuit current is given in Fig. 14b, which is revolved as we mentioned that is explained in the following.

4.4. Simulation on nail penetration test type 2 (without heat shield)

The terminal voltage data of lithium-ion cell in nail penetration tests type 2 (without heat shield) is fitted as a boundary condition in the numerical model. As can be seen from the Fig. 9 that the terminal voltage of the cell drops sharply in a few seconds at the moment of penetration. The electrically conductive nail conducts the positive and negative electrodes directly once it is penetrated, a very high short-circuit current flowing through the nail is generated instantly. If we neglect the chemical processes that might occur inside lithium-ion cell and assume the penetration-induced discharge process as a pure physical process, then the behavior of the cell is remarkably similar to the discharge behavior of a parallel-plate capacitor. Hence, the parallel-plate capacitor formula Eq. (9), current definition formula Eq. (10) and Ohm's Law Eq. (11) are employed to deduce the relationship between cell terminal voltage and time.

$$Q = C \cdot U \tag{9}$$

$$I(t) = \frac{dQ}{dt} \tag{10}$$

$$U(t) = I(t) \cdot R_{st} \tag{11}$$

$$-C \cdot \frac{dU}{dt} = \frac{U(t)}{R_{st}} \tag{12}$$

where Q(C) is the amount of charge stored in capacitor, C(F) is the capacitance of capacitor, U(V) is the voltage across capacitor, I(A) is current, $R_{st}(\Omega)$ is the shorting resistance.

Eq. (12) can be obtained by solving Eqs. (9), (10) and (11) simultaneously, the solution for this first order linear homogeneous ordinary differential equation is shown in Eq. (13).

$$U(t) = U(0) \cdot \exp\left(\frac{-t}{C \cdot R_{st}}\right) = U_0 \cdot \exp\left(\frac{-t}{C \cdot R_{st}}\right) \tag{13}$$

where U(t)(V) is cell terminal voltage, U_0(V) is the initial value of cell terminal voltage.

Substituting Eq. (14) into Eq. (13), the relationship between terminal voltage and time is given in Eq. (15), as can be seen that the terminal voltage does decay exponentially with time, where Q_0(C) is the initial charge amount stored in cell.

$$C = \frac{Q_0}{U_0} \tag{14}$$

$$U(t) = U_0 \cdot \exp\left(\frac{-U_0}{Q_0 \cdot R_{st}} \cdot t\right) \tag{15}$$

It can be concluded that the experimental terminal voltage can be evolved into an exponential decay function as the Eq. (15) displayed, thus the experimental terminal voltage and the corresponding exponentially fitted curves used in the model are demonstrated in Fig. 10. The value of root mean square error (RMSE) and R-Square (also known

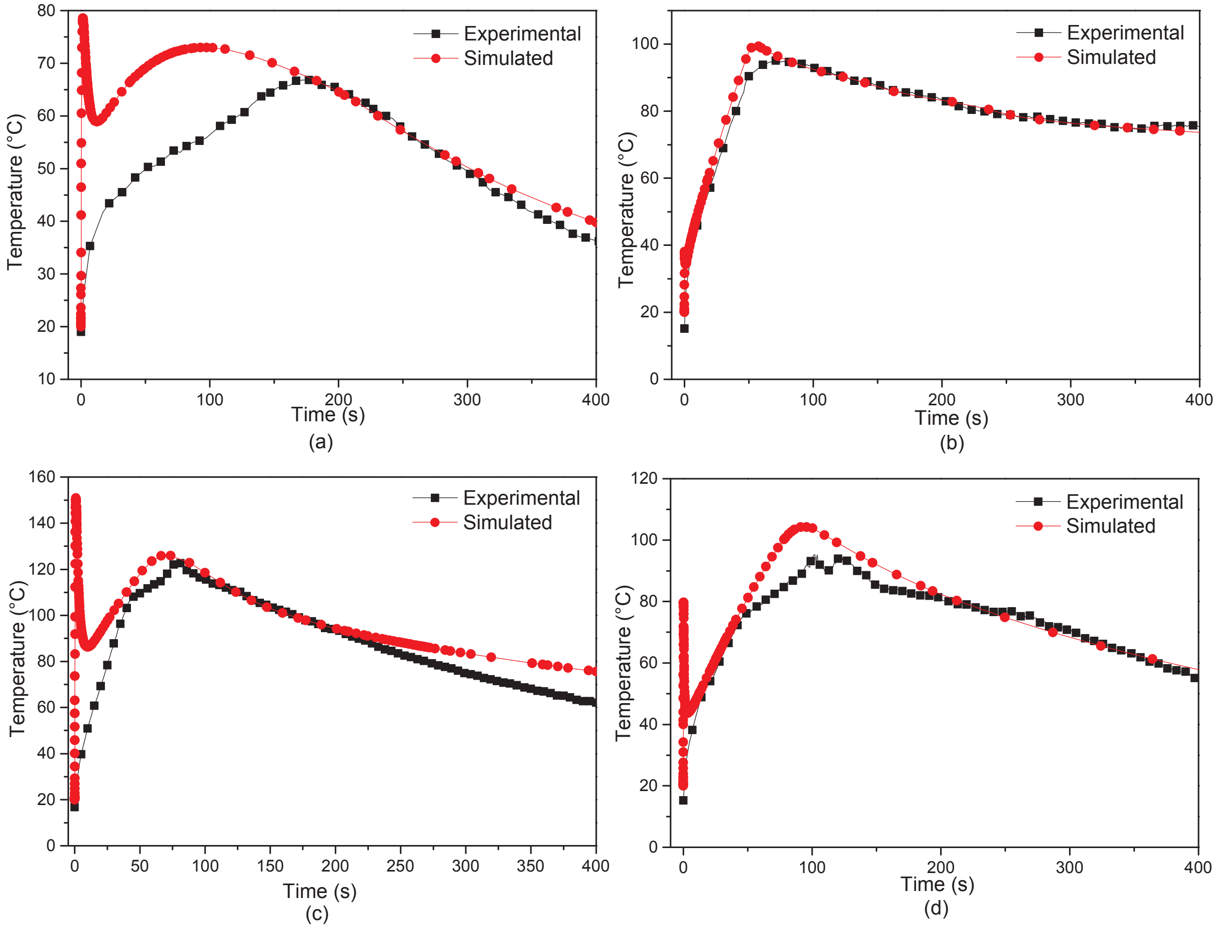

Fig. 11. Comparison between experimental temperature curves at nail site (TML2) for all experimental conditions and simulated results. (a) SOC of 50%, nail diameter of 5 mm; (b) SOC of 100%, nail diameter of 5 mm; (c) SOC of 100%, nail diameter of 3 mm; (d) SOC of 100%, nail diameter of 8 mm.

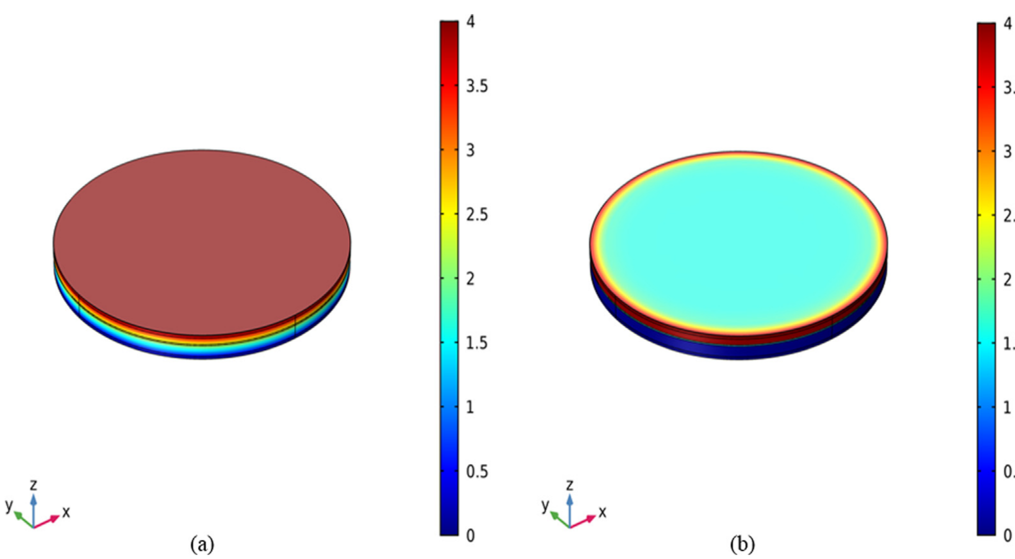

Fig. 12. The electric field distribution of (a) a common electrical conductor; and (b) a penetrated nail.

as coefficient of determination) of the fittings are also displayed, from which the fitted curves match well with the experimental data to ensure the accuracy of the model. It is worth mentioning that parallel plate capacitor hypothesis is only declared to explain the voltage curves, so that a more appropriate fitting function can be selected to fit the voltage curve. Moreover, the nail penetration model is only a tool to try to interpret some of the experimental results, but without claiming to be able to predict the behavior of cells in similar conditions.

The simulation procedure is carried out without heat shield to interpret some of the experimental results. The model geometry and parameters are shown in the Fig. 2 and Table 3. The temperature profiles at nail site of nail penetration tests type 2 and simulated results are

Table 5
Numerical values for nail resistance and fitted shorting resistance used in the model.

Experimental condition			Nail resistance $R_{nail}(m\Omega)$	Shorting resistance $R_{st}(m\Omega)$
SOC (%)	d_{nail} (mm)	v (mm s^{-1})		
50	5	20	6.0557e−4	39.12
100	5	20	6.0557e−4	212.36
100	3	20	1.6821e−3	71.50
100	8	20	2.3655e−4	69.69

displayed in Fig. 11, the simulation results show good agreement with the experimental results. It can be found that there is a difference between the two temperature curves in the early stage of temperature rise. A temperature spike upon initiation of the internal short-circuit is observed in simulation, while the experimental temperature does not. The spike in simulated temperature is because the internal short circuit is first occurred at the nail site, then the huge shorting current (as shown in Fig. 14b) caused the large Joule heat which dominates the total heat of the cell, so that forms a hot spot at the nail site. With the subsequent heat conduction and heat dissipation from the nail site, the temperature drops gradually. The difference between simulation and experiment has already been reported in Ref. [38] by comparing temperature profiles measured using Infra-Red imaging and thermocouple for a nail penetration test. The analogous lagging temperature at the surface of the cell can compared to the immediate rise of the nail is also observed in Ref. [39]. This phenomenon can be explained from two aspects: on the one hand, the response time of thermocouple is too long to capture the instantaneous temperature rise within millisecond range. While on the other hand, the heat conduction is delayed from the nail site to the surface of the cell, a large amount of heat was being conducted away from the short by the nail [39].

The shorting resistance in the model is discussed below. The shorting resistance is determined by the intrinsic resistance of the nail material and the contact resistance caused by the imperfect contact between the penetrated nail and the cell components, as shown in Eq. (16) [14]. Generally, the law of resistance is used to determine the resistance of an electrical conductor, as expressed in Eq. (17), the direction of electric field is usually perpendicular to the cross-section of the conductor, while the direction of electric field is not perpendicular to the cross-section of the nail in a nail penetration test, as can be seen in Fig. 12a and 12b, respectively. Different electric field distribution leads to different current path and thus resulting in different resistance of the nail. Therefore, the resistance of the penetrated nail cannot be directly calculated by Eq. (17).

$$R_{st} = R_{nail} + R_{ct} \tag{16}$$

$$R_{nail} = \frac{L_{pene}}{\sigma_{nail} A_{nail}} = \frac{4L_{pene}}{\pi \sigma_{nail} d_{nail}^2} \tag{17}$$

where R_{st}, R_{nail}, and$R_{ct}(\Omega)$ are the shorting resistance, penetrated nail resistance and contact resistance separately, $L_{pene}(m)$ is the penetration length which is the length of the nail portion that is inside the cell, that is the thickness of the nailed cell. $A_{nail}(m^2)$, σ_{nail} $(Sm^{-1})$and$d_{nail}(m)$ are the cross-sectional area, conductivity and diameter of the penetrated nail.

The contact resistance is also difficult to quantify and is a complex function with nail diameter, penetration speed, cell and nail material

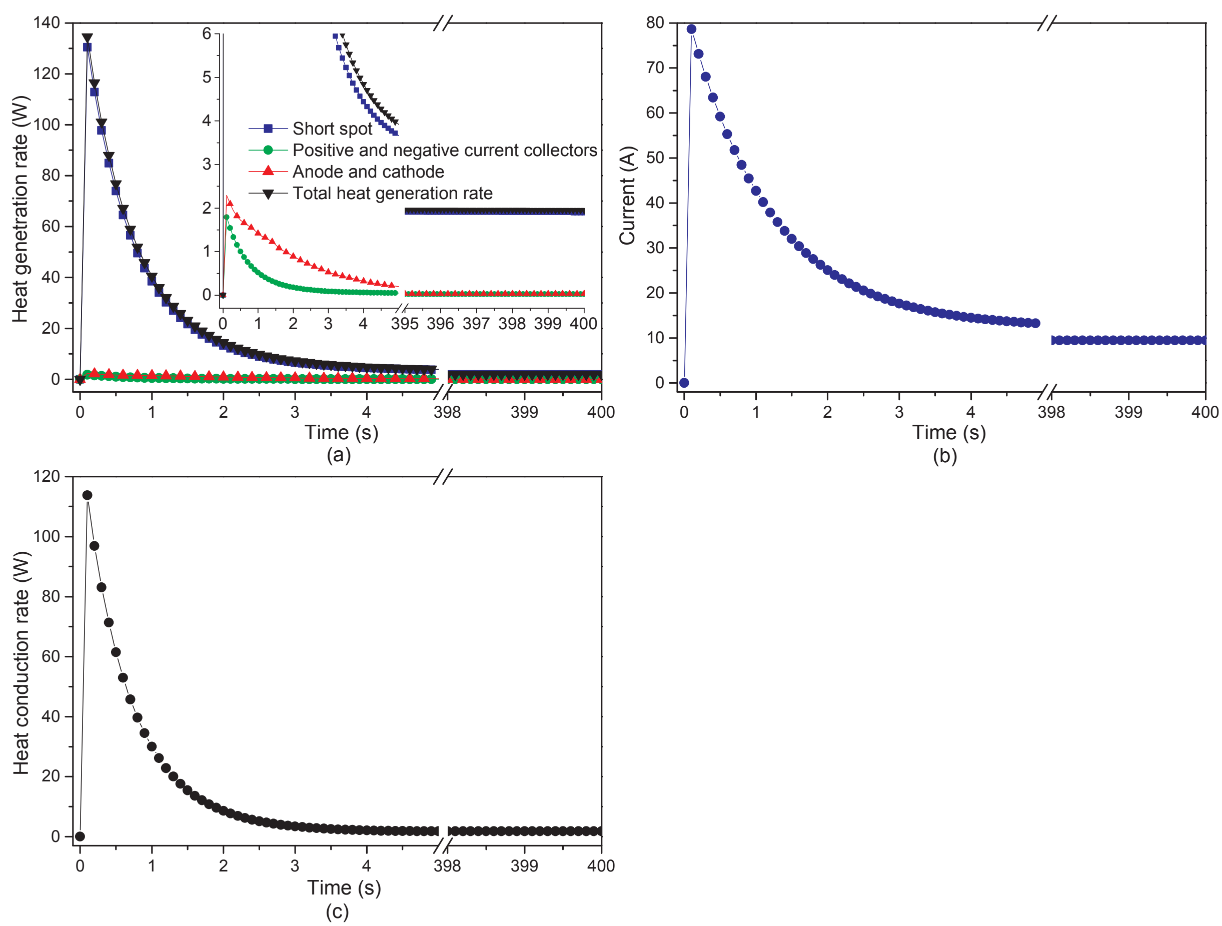

Fig. 13. Numerical results of (a) heat generation rate of each component inside lithium-ion cell; (b) current through nail cross-section; and (c) heat conduction rate from short spot to nail at SOC of 100%, nail diameter of 3 mm during the nail penetration.

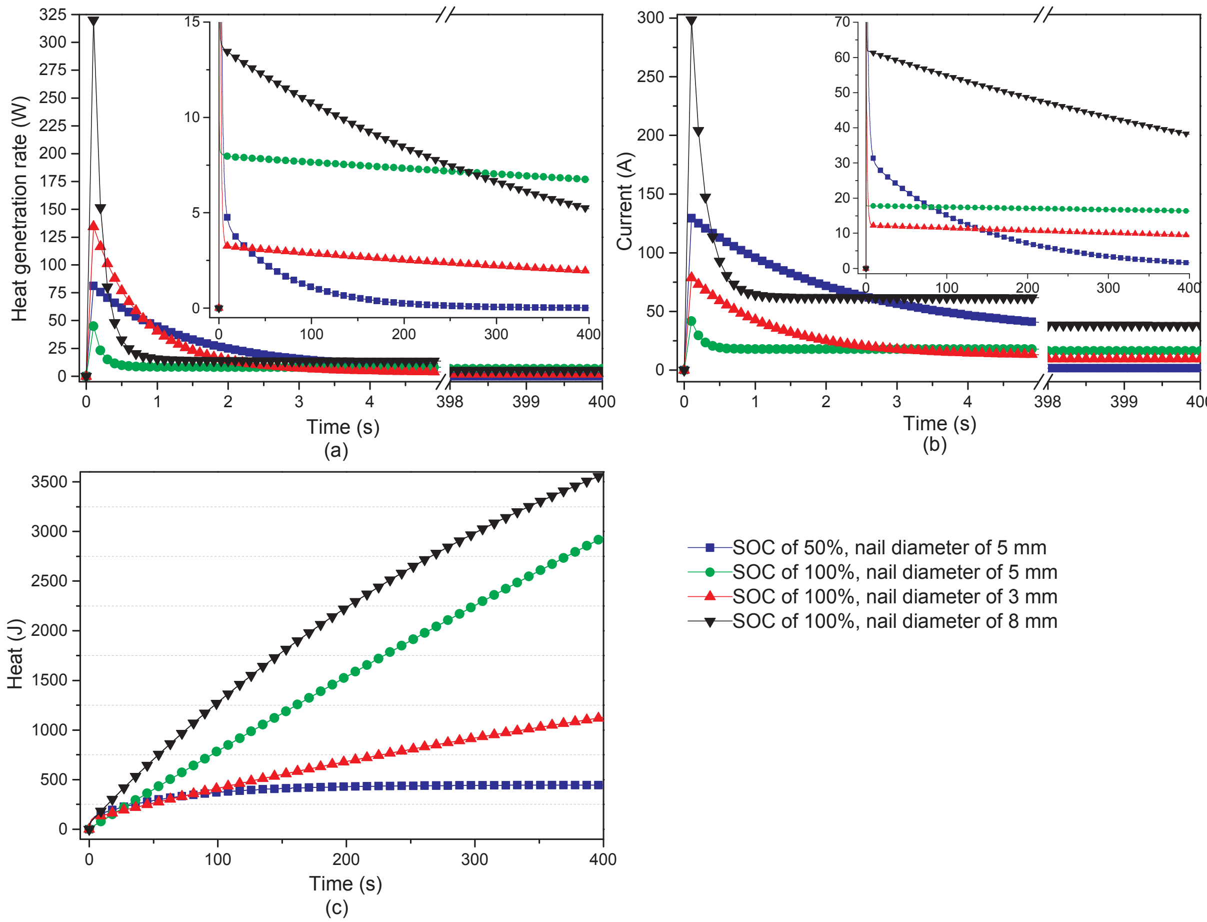

Fig. 14. Numerical results of (a) total heat generation rate inside the cell; (b) current through nail cross-section; and (c) gross heat generated inside the cell profiles during the nail penetration.

properties, etc. To date, there is few experimental researches aiming at measuring the contact resistance or providing clues on how to precisely control the contact resistance during the nail penetration process [14]. Considering the difficulty to obtain the analytical solution of the resistance of a penetrated nail, the numerical solution of the equivalent shorting resistance is obtained using finite element method by the electrochemical-thermal model. The shorting current passing through the nail body will be calculated according to the electrochemical model, the equivalent shorting resistance is obtained by dividing the potential gradient by shorting current, as shown in Eq. (18) [14],

$$R_{st} = \frac{\Delta\phi_s}{I_s} \tag{18}$$

where $\Delta\phi_s$ is the solid potential gradient along the nail axial direction, I_s is the shorting current.

Finally, the equivalent shorting resistance of penetrated nail is used as a fitting parameter in order to match the simulation with the experiment. The calculated values for penetrated nail resistance and fitted shorting resistance used in the model are listed in Table 5, it can be found that the shorting resistance is several orders of magnitude larger than the penetrated nail resistance, which means the shorting resistance is mainly contributed by the contact resistance caused by the imperfect contact between the penetrated nail and cell components.

The temperature rising during the nail penetration process is due to various heat contribution inside the cell and is greatly affected by the heat conduction effect of the nail. The simulated heat generation rate of each component inside cell, current passing through nail cross-section and the heat conduction rate from short spot to nail (SOC of 100%, nail diameter of 3 mm) during the nail penetration are exhibited in Fig. 13a, b, and c, respectively. As illustrated in Fig. 13a, the total heat generation inside cell is mainly contributed by the Joule heat generated at short spot throughout the whole nail penetration process, while the Joule heat generated at positive and negative current collectors and the Joule heat, polarization heat and entropy heat generated at anode and cathode contribute only a very small part and quickly reduce to almost zero within the first few seconds after the internal short-circuit, which is traced back to the following two reasons. Firstly, Al-Cu short is the main shorting mode because the steel nail directly connects Al and Cu current collectors. Secondly, the shorting resistance at short spot is larger than the resistance of Al and Cu current collectors, indicating that most heat will be generated at the short spot in case of the same current. The sudden drop of Joule heat at short spot is aroused by the sharp decrease of short-circuit current as shown in Fig. 13b. The heat conduction rate from short spot to the nail is displayed in Fig. 13c, which resembles the Fig. 13b, it can be seen that a large portion of heat is conducted to the nail compared to the total heat generation rate inside cell shown in Fig. 13a.

The simulated total heat generation rate inside cell, current through nail cross-section and gross heat generated profiles with the four cases during the nail penetration are displayed in Fig. 14a, b, and c, respectively. The gross heat profiles shown in Fig. 14c are obtained from the integration of total heat generation rate with respect to time. Nail with a larger diameter could provide a larger short-circuit current (shown in Fig. 14b), which consequently provides a greater total heat generation rate (shown in Fig. 14a) and a greater gross heat (shown in Fig. 14c). However, it can also be found from the Fig. 14a and b that the initial heat generation rate and current spike at the SOC of 100% and nail diameter of 5 mm is the lowest, which further demonstrates the slight

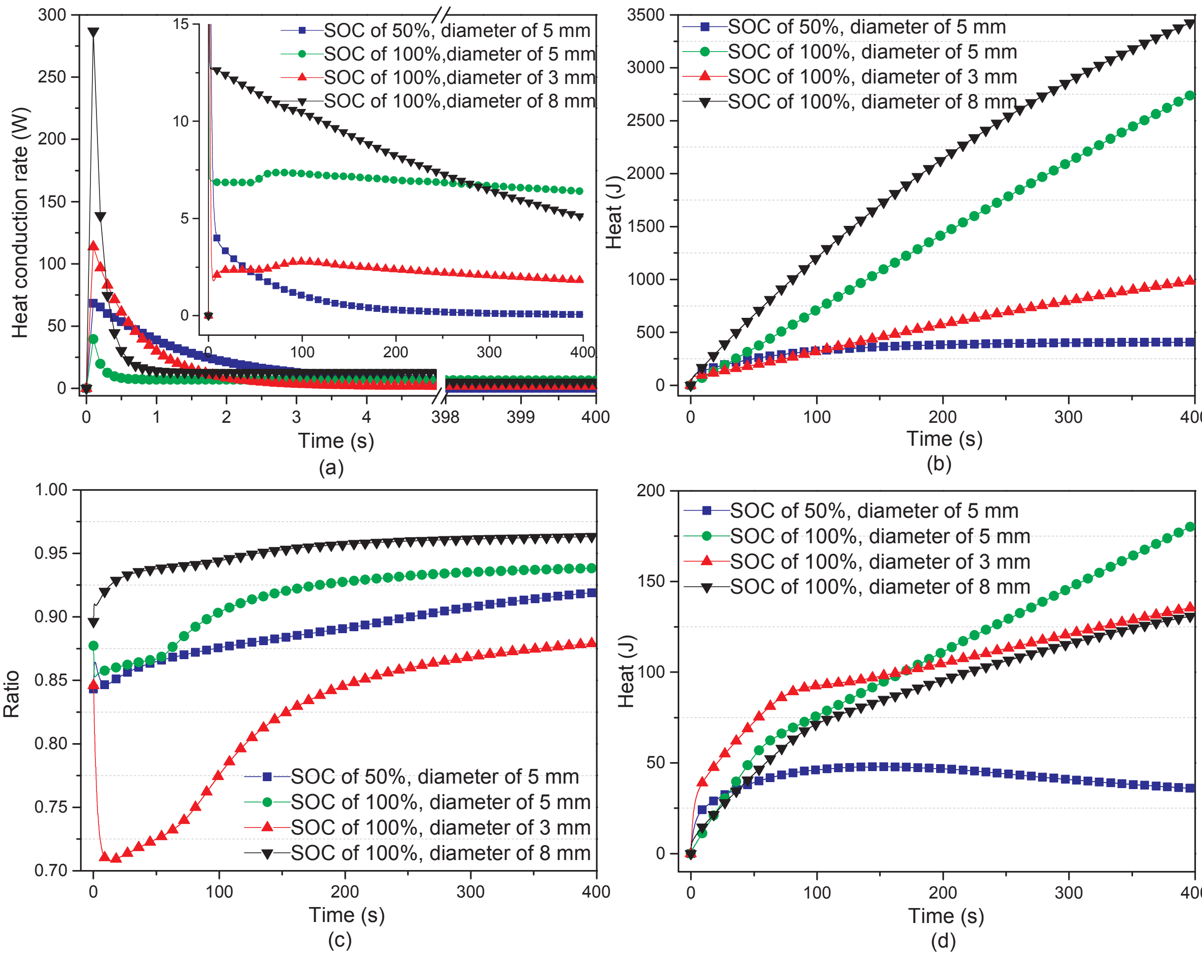

Fig. 15. Numerical results of (a) the heat conduction rate from short spot to nail; (b) gross heat conducted to the nail; (c) ratio of gross heat conducted to nail to the gross heat generated inside the cell; and (d) net heat absorbed by cell body *vs*. time plots during the nail penetration.

temperature sharp in Fig. 11b. It can also be seen that cell with a higher SOC displays a greater total heat generation rate.

Fig. 15a illustrates the simulated heat conduction rate curves from short spot to the nail, the gross heat conducted to nail (shown in Fig. 15b) are obtained from the integration of heat conduction rate to nail with respect to time. The ratio of gross heat conducted to nail to the gross heat generated inside the cell is displayed in Fig. 15c, it can be seen that a large proportion of heat (over 70%) is conducted to the nail and the ratio increases as nail diameter increases. It can be revealed from Fig. 15a, b and c that nail with a larger diameter has a greater ability to conduct heat from short spot. The temperature rise during the nail penetration process is determined by the net heat absorbed by the cell body, which can be obtained by subtracting the gross heat conducted to nail from the gross heat generated inside the cell since the amount of heat dissipated through heat convection and radiation is quite small when compared to gross heat conducted to the nail as shown in Fig. 15d. The temperature curves of the nail penetration tests type 2 (without heat shield) in Figs. 8, 9 and 11 can be perfectly explained using Fig. 15d. The experiment results show that temperature rising rate and the maximum temperature at nail site both increase as the SOC increases and as the nail diameter decreases (shown in Fig. 8a), which is because the net heat absorbed by cell body increase as the SOC increases and as the nail diameter decreases (as shown in Fig. 15d).

Fig. 16 demonstrates the simulated 3D Celsius scale temperature diagrams at various times with SOC of 50% and nail diameter of 5 mm. As shown in Fig. 16a, b, and c, heat propagates in all directions from the short-circuit area after nail penetration and the highest temperature is located at nail site in the early stage of penetration.

5. Conclusions

In this study, the penetration-induced internal short-circuit of lithium-ion cell is investigated through a combination of nail penetration experiments and numerical simulation, the model predictions agree well with the nail penetration test results. In addition, cell separator thermal analysis is carried out. The main conclusions are as follows:

(1) The temperature rising rate and maximum temperature of the cell both increase with the increase of SOC, which is due to a higher SOC could provide a larger short-circuit current during the nail penetration process.
(2) Two cases with thermal runaway and non-thermal runaway are observed during the nail penetration test. The nail plays a dual role in the penetration process: In case of thermal runaway, nail with a larger diameter provides a larger short-circuit current, which consequently generates a greater total heat generation rate and thus takes less time to trigger thermal runaway. While in case of non-thermal runaway, a larger nail diameter provides larger shorting area, displaying a greater ability in heat dissipation.
(3) The temperature rise modes for thermal runaway cases are divided into two typical forms with the sharp temperature rise and gradual temperature rise.
(4) The cell terminal voltage is found to decay exponentially with time after penetration, while the fluctuations and rebound of terminal voltage during the nail penetration process are attributed to the partially interrupted internal short-circuit path.
(5) The Joule heat generated at short spot dominates the total heat

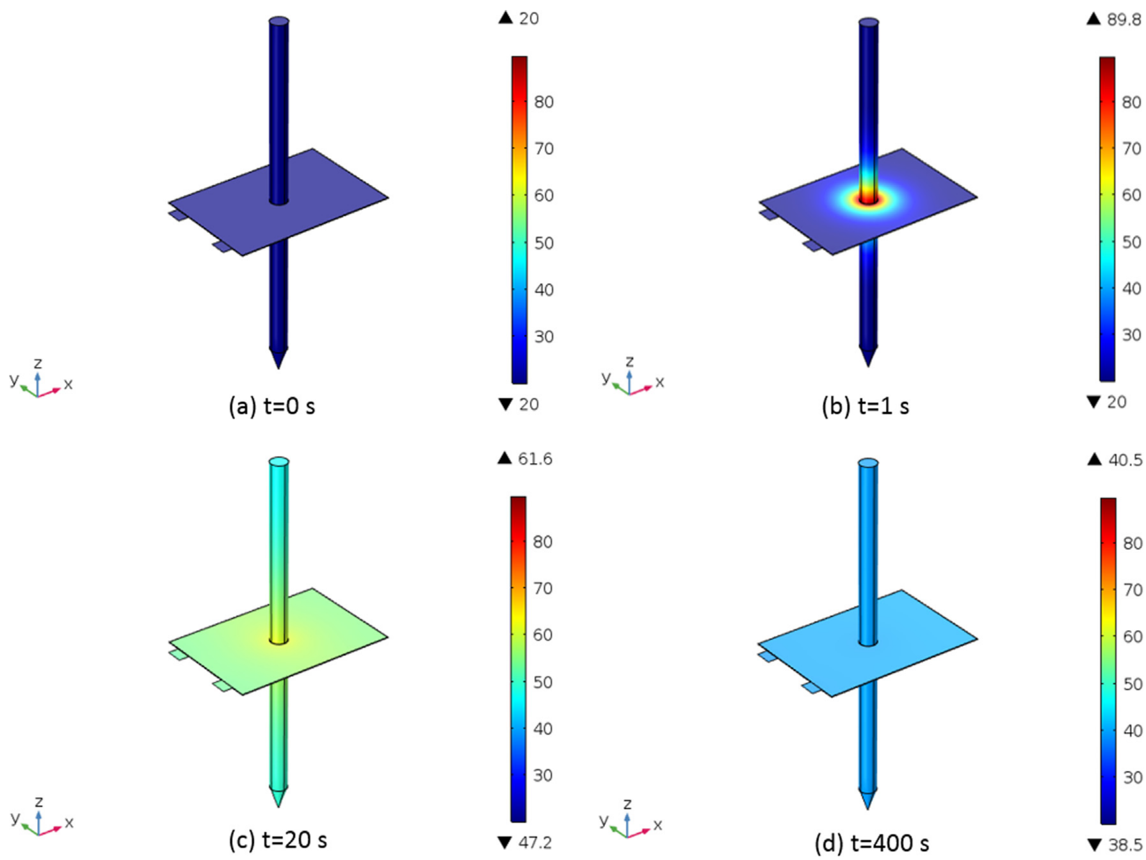

Fig. 16. Simulated 3D Celsius scale temperature diagrams (unit: °C) at various times with SOC of 50%, nail diameter of 5 mm. (a) t = 0 s; (b) t = 1 s; (c) t = 20 s; (d) t = 400 s.

generation inside lithium-ion cell with non-thermal runaway nail penetration cases.

The model in our study is more to explain the experimental phenomenon and analyze the mechanism of nail penetration caused internal short circuit, more sophisticated internal short-circuit models can be proposed on the basis of this work in the future.

Declaration of Competing Interest

None.

Acknowledgements

This work is supported by the National Key R&D Program of China (No. 2016YFB0100306), the National Natural Science Foundation of China (No. 51674228 & 51976209), and the Fundamental Research Funds for the Central Universities (No. WK2320000038). Dr. Q.S. Wang is supported by Youth Innovation Promotion Association CAS (No.2013286).

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.applthermaleng.2020.115082.

References

[1] Qingsong Wang, Ping Ping, Xuejuan Zhao, Guanquan Chu, Jinhua Sun, Chunhua Chen, Thermal runaway caused fire and explosion of lithium ion battery, J. Power Sources 208 (2012) 210–224.
[2] X.N. Feng, X.M. He, M.G. Ouyang, L.G. Lu, P. Wu, C. Kulp, S. Prasser, Thermal runaway propagation model for designing a safer battery pack with 25 Ah LiNixCoyMnzO2 large format lithium ion battery, Appl. Energy 154 (2015) 74–91.
[3] Ping Ping, Qingsong Wang, Peifeng Huang, Jinhua Sun, Chunhua Chen, Thermal behaviour analysis of lithium-ion battery at elevated temperature using deconvolution method, Appl. Energy 129 (2014) 261–273.
[4] D. Miranda, C.M. Costa, A.M. Almeida, S. Lanceros-Méndez, Computer simulations of the influence of geometry in the performance of conventional and unconventional lithium-ion batteries, Appl. Energy 165 (2016) 318–328.
[5] Binbin Mao, Haodong Chen, Zhixian Cui, Wu. Tangqin, Qingsong Wang, Failure mechanism of the lithium ion battery during nail penetration, Int. J. Heat Mass Transf. 122 (2018) 1103–1115.
[6] Joshua Lamb, Christopher J. Orendorff, Evaluation of mechanical abuse techniques in lithium ion batteries, J. Power Sources 247 (2014) 189–196.
[7] Daniel H. Doughty, Chris C. Crafts, FreedomCAR: Electrical Energy Storage System Abuse Test Manual for Electric and Hybrid Electric Vehicle Applications, Sandia National Laboratories, 2006.
[8] Hossein Maleki, Jason N. Howard, Internal short circuit in Li-ion cells, J. Power Sources 191 (2) (2009) 568–574.
[9] T. Yokoshima, D. Mukoyama, F. Maeda, T. Osaka, K. Takazawa, S. Egusa, Operando analysis of thermal runaway in lithium ion battery during nail-penetration test using an X-ray inspection system, J. Electrochem. Soc. 166 (6) (2019) A1243–A1250.
[10] T. Yokoshima, D. Mukoyama, F. Maeda, T. Osaka, K. Takazawa, S. Egusa, S. Naoi, S. Ishikura, K. Yamamoto, Direct observation of internal state of thermal runaway in lithium ion battery during nail-penetration test, J. Power Sources 393 (2018) 67–74.
[11] Donal P. Finegan, Bernhard Tjaden, Thomas M. M. Heenan, Rhodri Jervis, Marco Di Michiel, Alexander Rack, Gareth Hinds, Dan J. L. Brett, Paul R. Shearing, Tracking internal temperature and structural dynamics during nail penetration of lithium-ion cells, J. Electrochem. Soc. 164(13) (2017) A3285-A3291.
[12] Rui Zhao, Jie Liu, Gu. Junjie, A comprehensive study on Li-ion battery nail penetrations and the possible solutions, Energy 123 (2017) 392–401.
[13] Guozhou Liang, Yiming Zhang, Qi Han, Zhaoping Liu, Zhen Jiang, Shuang Tian, A novel 3D-layered electrochemical-thermal coupled model strategy for the nail-penetration process simulation, J. Power Sources 342 (2017) 836–845.
[14] W. Zhao, G. Luo, C.Y. Wang, Modeling nail penetration process in large-format Li-Ion cells, J. Electrochem. Soc. 162 (1) (2015) A207–A217.
[15] Shriram Santhanagopalan, Premanand Ramadass, John Zhang, Analysis of internal short-circuit in a lithium ion cell, J. Power Sources 194 (1) (2009) 550–557.
[16] T.G. Zavalis, M. Behm, G. Lindbergh, Investigation of short-circuit scenarios in a lithium-ion battery cell, J. Electrochem. Soc. 159 (6) (2012) A848–A859.

[17] Weifeng Fang, Premanand Ramadass, Zhengming Zhang, Study of internal short in a Li-ion cell-II. Numerical investigation using a 3D electrochemical-thermal model, J. Power Sources 248 (2014) 1090–1098.
[18] Xuning Feng, Caihao Weng, Minggao Ouyang, Jing Sun, Online internal short circuit detection for a large format lithium ion battery, Appl. Energy 161 (2016) 168–180.
[19] Rui Zhao, Jie Liu, Gu. Junjie, Simulation and experimental study on lithium ion battery short circuit, Appl. Energy 173 (2016) 29–39.
[20] Xu. Jun, Wu. Yijing, Sha Yin, Investigation of effects of design parameters on the internal short-circuit in cylindrical lithium-ion batteries, RSC Adv. 7 (24) (2017) 14360–14371.
[21] Daniel J. Noelle, Meng Wang, Anh V. Le, Yang Shi, Yu Qiao, Internal resistance and polarization dynamics of lithium-ion batteries upon internal shorting, Appl. Energy 212 (2018) 796–808.
[22] Wenxin Mei, Qiangling Duan, Chunpeng Zhao, Wei Lu, Jinhua Sun, Qingsong Wang, Three-dimensional layered electrochemical-thermal model for a lithium-ion pouch cell Part II. The effect of units number on the performance under adiabatic condition during the discharge, Int. J. Heat Mass Transf. 148 (2020) 119082.
[23] Andreas Nyman, Mårten Behm, Göran Lindbergh, Electrochemical characterisation and modelling of the mass transport phenomena in LiPF6–EC–EMC electrolyte, Electrochim. Acta 53 (22) (2008) 6356–6365.
[24] Karen E. Thomas, John Newman, Heats of mixing and of entropy in porous insertion electrodes, J. Power Sources 119 (2003) 844–849.
[25] Chao Zhang, Shriram Santhanagopalan, Michael A. Sprague, Ahmad A. Pesaran, Coupled mechanical-electrical-thermal modeling for short-circuit prediction in a lithium-ion cell under mechanical abuse, J. Power Sources 290 (2015) 102–113.
[26] COMSOL Material Library (2016). COMSOL multiphysics, v5.2a. MA, USA: Burlington.
[27] John Newman, William Tiedemann, Porous-electrode theory with battery applications, AIChE J. 21 (1975) 25–41.
[28] Kuan-Cheng Chiu, Chi-Hao Lin, Sheng-Fa Yeh, Yu-Han Lin, Kuo-Ching Chen, An electrochemical modeling of lithium-ion battery nail penetration, J. Power Sources 251 (2014) 254–263.
[29] John Newman, Karen E. Thomas, Hooman Hafezi, Dean R. Wheeler, Modeling of lithium-ion batteries, J. Power Sources 119–121 (2003) 838–843.
[30] Kandler Smith, Chao-Yang Wang, Power and thermal characterization of a lithium-ion battery pack for hybrid-electric vehicles, J. Power Sources 160 (1) (2006) 662–673.
[31] Yonghuang Ye, Yixiang Shi, Ningsheng Cai, Jianjun Lee, Xiangming He, Electro-thermal modeling and experimental validation for lithium ion battery, J. Power Sources 199 (2012) 227–238.
[32] R. Spotnitz, J. Franklin, Abuse behavior of high-power, lithium-ion cells, J. Power Sources 113 (1) (2003) 81–100.
[33] M.N. Richard, J.R. Dahn, Accelerating rate calorimetry study on the thermal stability of lithium intercalated graphite in electrolyte. I. Experimental, J. Electrochem. Soc. 146 (6) (1999) 2068–2077.
[34] Xuning Feng, Mou Fang, Xiangming He, Minggao Ouyang, Lu. Languang, Hao Wang, Mingxuan Zhang, Thermal runaway features of large format prismatic lithium ion battery using extended volume accelerating rate calorimetry, J. Power Sources 255 (2014) 294–301.
[35] Haiyan Wang, Aidong Tang, Kelong Huang, oxygen evolution in overcharged LixNi1/3Co1/3Mn1/3O2 electrode and its thermal analysis kinetics, Chin. J. Chem. 29 (8) (2011) 1583–1588.
[36] M.S. Wu, P.C.J. Chiang, J.C. Lin, Y.S. Jan, Correlation between electrochemical characteristics and thermal stability of advanced lithium-ion batteries in abuse tests - short-circuit tests, Electrochim. Acta 49 (11) (2004) 1803–1812.
[37] Xiaowei Zhang, Elham Sahraei, Kai Wang, Li-ion battery separators, mechanical integrity and failure mechanisms leading to soft and hard internal shorts, Sci. Rep. 6 (2016).
[38] Premanand Ramadass, Weifeng Fang, Zhengming Zhang, Study of internal short in a Li-ion cell I. Test method development using infra-red imaging technique, J Power Sources 248 (2014) 769–776.
[39] T.D. Hatchard, S. Trussler, J.R. Dahn, Building a “smart nail” for penetration tests on Li-ion cells, J. Power Sources 247 (2014) 821–823.